

GRANT COUNTY INTL, WA, USA

WMO: 727827

Lat: 47.208N

Lon: 119.319W

Elev: 1196

StdP: 14.07

Time Zone: -8.00 (NAP)

Period: 97-19

WBAN: 24110

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	7.2	12.6	-0.2	5.7	10.2	5.0	7.4	16.1	24.7	38.3	21.3	38.9	5.7	350	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.5	98.2	66.5	94.8	64.6	91.2	64.0	69.0	92.0	66.9	89.7	65.2	88.0	8.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	83.8	77.7	57.8	74.5	74.8	55.5	68.6	73.1	33.7	91.2	32.0	90.1	30.6	88.0	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.3	18.8	16.5	DB	0.4	103.5	8.9	3.0	-6.1	105.7	-11.3	107.4	-16.3	109.1	-22.8	111.3	
			WB	-0.3	71.5	8.8	3.8	-6.7	74.2	-11.8	76.5	-16.8	78.7	-23.2	81.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	30.4	35.6	43.6	50.7	59.7	66.7	74.2	72.8	63.3	50.3	38.1	29.7	(6)
	DBStd	16.85	8.42	7.06	6.32	5.97	6.94	6.57	5.76	5.51	6.35	6.71	8.41	8.39	(7)
	HDD50	2357	607	405	209	61	6	0	0	0	2	76	362	629	(8)
	HDD65	5705	1072	824	662	429	193	55	4	6	105	456	806	1093	(9)
	CDD50	2853	0	1	12	82	308	500	751	706	400	86	6	1	(10)
	CDD65	726	0	0	0	1	30	105	289	247	53	1	0	0	(11)
	CDH74	11612	0	0	0	57	634	1692	4466	3700	1035	28	0	0	(12)
	CDH80	5631	0	0	0	9	211	721	2410	1889	387	4	0	0	(13)
(14) Wind	WSAvg	7.0	5.8	6.8	7.8	8.3	8.1	8.2	7.4	6.8	6.7	6.7	6.4	5.7	(14)
(15) Precipitation	PrecAvg	7.1	0.9	0.5	0.7	0.5	0.6	0.7	0.2	0.2	0.2	0.7	0.9	1.1	(15)
	PrecMax	10.2	2.6	1.1	1.6	1.2	1.7	1.6	0.7	0.8	0.7	2.5	1.8	2.5	(16)
	PrecMin	4.5	0.2	0.0	0.2	0.0	0.2	0.0	0.0	0.0	0.0	0.1	0.1	0.1	(17)
	PrecStd	1.9	0.7	0.3	0.4	0.3	0.5	0.5	0.2	0.2	0.2	0.6	0.4	0.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.1	57.4	67.7	80.5	91.2	99.3	102.3	100.4	94.5	78.4	62.5	54.9	(19)
		MCWB	45.0	48.0	52.5	57.6	61.4	67.5	67.5	66.9	64.3	59.1	52.9	47.2	(20)
	2%	DB	47.5	52.6	63.3	73.8	85.9	92.5	98.9	97.4	89.0	72.0	57.2	48.2	(21)
		MCWB	41.5	44.3	49.7	54.6	60.2	65.1	67.3	65.9	63.4	55.5	48.5	43.2	(22)
	5%	DB	43.9	49.5	59.5	69.9	81.1	88.0	95.6	93.4	84.1	67.5	53.8	44.6	(23)
		MCWB	39.5	42.5	47.2	52.6	58.5	62.9	64.7	64.5	60.7	52.6	46.8	40.4	(24)
	10%	DB	40.1	46.1	56.1	64.8	76.5	83.4	92.4	90.3	80.4	63.6	50.3	40.0	(25)
		MCWB	37.1	40.4	45.3	49.8	56.4	60.5	64.3	63.7	59.3	50.9	44.6	37.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	49.2	54.0	59.2	64.0	71.1	73.0	71.5	68.1	60.7	53.9	48.5	(27)
		MCDB	50.5	56.2	64.8	76.9	86.7	88.6	97.4	96.3	89.6	73.7	59.9	55.0	(28)
	2%	WB	42.3	45.6	50.7	55.7	61.6	66.5	69.4	68.1	64.4	56.9	50.3	43.9	(29)
		MCDB	46.5	51.0	60.7	71.6	81.6	89.4	92.7	91.4	84.9	68.0	55.7	47.9	(30)
	5%	WB	39.9	43.3	48.8	53.2	59.7	64.0	67.4	66.2	62.2	54.5	47.8	40.6	(31)
		MCDB	43.3	48.3	57.9	67.7	77.4	85.9	90.3	89.0	80.9	64.8	52.3	43.9	(32)
	10%	WB	37.4	41.1	46.6	50.8	57.8	61.7	65.5	64.5	60.1	52.4	45.5	37.4	(33)
		MCDB	39.5	45.6	54.6	63.1	73.4	80.4	88.4	86.5	77.0	61.7	49.3	39.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	16.1	21.4	24.7	26.2	26.4	30.5	30.5	29.2	23.7	16.4	11.5	(35)
		MCDBR	15.4	19.8	26.4	32.2	33.4	33.9	35.6	35.9	36.0	28.7	19.9	15.4	(36)
		MCWBR	11.1	12.8	15.1	16.0	13.3	13.7	13.3	13.8	16.2	15.4	13.0	11.3	(37)
	5% WB	MCDBR	14.3	17.9	24.6	30.7	29.6	32.3	32.4	32.4	32.4	24.5	17.7	14.1	(38)
		MCWBR	10.7	12.2	14.9	15.8	12.6	13.7	13.7	13.7	15.5	14.0	12.3	10.9	(39)
(40) Clear-Sky Solar Irradiance	taub		0.278	0.282	0.293	0.321	0.335	0.336	0.337	0.337	0.323	0.303	0.287	0.273	(40)
	taud		2.528	2.573	2.573	2.460	2.437	2.462	2.455	2.447	2.485	2.543	2.547	2.526	(41)
	Ebn at Noon		264	287	299	297	295	294	292	288	283	274	257	252	(42)
	Edh at Noon		19	23	27	33	35	34	34	33	29	23	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg		345	703	1121	1661	2024	2224	2408	2015	1485	914	460	286	(44)
	RadStd		41	89	77	100	123	122	91	109	82	74	50	37	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	+1.66	N/A	N/A	N/A	N/A	+225	+126
Regional (19 neighbors)	N/A	N/A	N/A	N/A	-0.65	N/A	N/A	N/A	+176	+88

Nomenclature: See separate page